Homework 2 Due on 3/18/2026

Note:

- You may need to use Matlab or other software for this homework.
- You are not allowed to copy from others' solution or any solution on line.

1. Thermal radiation

We briefly learned about black-body radiation in class. A simple way to understand this is that any object with non-zero temperature (T, unit: Kelvin) will radiate EM-wave (thermal radiation). The black body spectral radiance (power per emitting area per solid angle per unit wavelength) is given by

$$I(\lambda, T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda kT}} - 1} \quad (1)$$

where h = 6.62×10^{-34} m^2kg/s is Plank constant, k = 1.38×10^{-23} $m^2kg/s^2/K$ is Boltzmann constant and c is speed of light.

a) At a certain temperature T, there exists a wavelength λ_m that has the strongest emission intensity according to equation (1). Show that λ_m satisfies this equation: $hc/(\lambda kT) = 5(1 - \exp(-hc/\lambda kT))$. (Hint: same as finding the maximum of a function in calculus.) (10pts)

b) One solution of this equation is $hc/(\lambda kT) = 0$. Can you use Mathematica/Matlab to find the other numerical value of $hc/(\lambda kT)$ that satisfies the equation above in (a)? (5pts)

c) We can write the solution in (b) as $\lambda m = b/T$. What is the numerical value of constant b? This equation is also called Wien's displacement law. It was discovered experimentally before Plank's black body radiation equation. (5pts)

d) The radiation spectrum of our sun has maximum emission at wavelength 500 nm. What is the corresponding temperature of the sun? (This corresponds to the surface temperature of the sun) (5pts) $\lambda_m = \frac{b}{T}$

e) Google (or Baidu) sun's surface temperature. Is it close to your result? Fun fact: This is how people determine the temperature of star. (5pts)

f) What frequency has the strongest emission at room temperature (300K)? (5pts)

2. Quantum well

Quantum well can be considered as man-made atoms/molecules. It has a series of energy levels. Unlike nature atoms/molecules, the energy levels of quantum well can be controlled by engineering the physical dimensions of the quantum well. Quantum well is now widely used in lasers and optical amplifiers. The figure below shows a simplified configuration of one-dimensional quantum well. An electron is trapped in a potential well with infinite potential barriers.

a) If electron is wave, what boundary condition does the wave satisfy in the quantum well? (5pts) $\psi(0) = \psi(L) = 0$

b) Using the boundary condition, what are the possible wavelengths of the wave? You can

$\lambda = \frac{2L}{n}$ $\sin kx + \cos kx$ $kL = n\pi$

assume the wave has the form of sine or cosine wave (not necessary, but it will simplify the analysis). (5pts)

c) What is the corresponding wavenumber (k) for these wavelengths? (5pts)

d) The energy of the electron can be expressed as $E = (\hbar k)^2/2m$. What are the energy levels for the electron? (5pts) $k = \frac{\sqrt{2mE}}{\hbar}$ $k = \frac{n\pi}{L}$ $E = \frac{n^2\pi^2\hbar^2}{2mL^2}$

e) Usually, people excite the electron in the quantum well into its second lowest energy state (first excited state), and the electron will decay to the lowest energy state (ground state), and emit a photon. Suppose we want the color of the photon to be green (photon wavelength equals to 500 nm), what should be the length of quantum well (L)? You will need to look up the mass of the electron. (5pts)

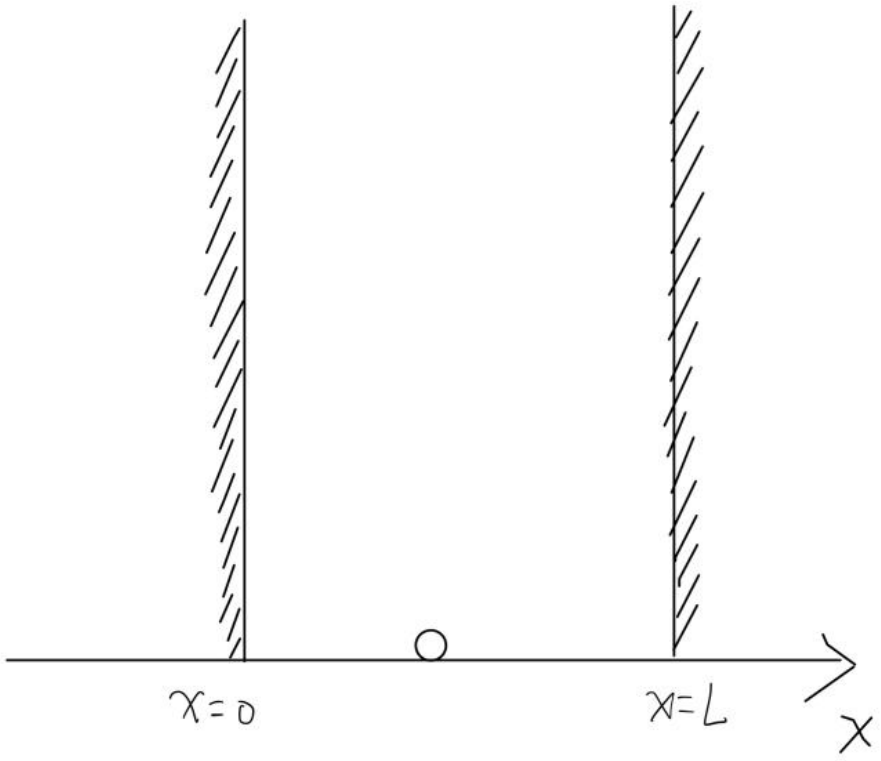


Figure 1: One dimensional quantum well.

3. De Broglie wavelength

I want everyone to calculate your own De Broglie wavelength, but I can't ask you to submit the answer, as this will clearly offend your privacy. So let's do it to our imaginary friend, Ted. Ted weights 50 kg, and he is good at sprinting. He can run up to 10 m/s (close to Olympic championing level).

a) What is the De Broglie wavelength of Ted when he sprints? (5pts) $p = \frac{h}{\lambda}$

b) Suppose Ted wants to use himself see some quantum effect, for example, to do double slit interference experiment. The double slit experiment would require the slit to be narrower than the wavelength. Do you think it's possible to build such a narrow slit? (5pts)

c) Now Ted took your advice and use electron, instead of himself for the double slit interference experiment. He adds a setup to accelerate the electron by connecting the 220 Volt standard electric outlet to a pair of parallel metal sheets. What is the momentum of the electron after acceleration? What is the De Broglie wavelength? (5pts)

d) If Ted wants to create electron interference pattern that is the same as the pattern of light at 500 nm, what voltage he should apply to accelerate his electrons? (5pts)

e) Do you think he can do this experiment at room temperature? Hint: At room temperature, on average each electron will have kinetic energy of 1.5kT, where k = 1.38×10^{-23} $m^2kg/s^2/K$ is Boltzmann constant, T is temperature in Kelvin. This kinetic energy will cause random motion of the electron, and can be considered as a noise source for the experiment. (5pts)

4. Considering the electron in 1D infinite quantum well (1D box). (15pts)

$$V(x) = \begin{cases} 0, & 0 < x < a \\ +\infty, & x \geq a \text{ or } x \leq 0 \end{cases}$$

a) please calculate <p_x> for all the energy states

b) please verify the normalization factor for wavefunctions is $\sqrt{\frac{2}{a}}$.

c) please calculate <x> for all the energy states.

$\psi(x) = \sqrt{\frac{2}{a}} \cdot \sin kx \qquad k = \frac{\sqrt{2mE}}{\hbar}$

$\frac{e^{jkx} - e^{-jkx}}{2} \cdot \frac{e^{jkx} + e^{-jkx}}{2} \cdot k$

$\frac{e^{jkx} - e^{-jkx}}{4} k$

$\langle p_x \rangle = \int_0^a \psi(x) \cdot (-i\hbar \frac{\partial}{\partial x}) \psi^*(x)\, dx$

$\frac{2}{a} \cdot (-\hbar) \cdot \int_0^a \frac{\sin 2kx}{2} \cdot k\, dx = 0$

$\frac{2}{a} \int_0^a x \cdot \sin^2 kx\, dx$

$\frac{2}{a} \int_0^a x \frac{1 - \cos 2kx}{2} dx = \frac{2}{a} \int_0^a \frac{x}{2} dx - \frac{1}{a} \int x \cos 2kx\, dx$

$= \frac{a}{2}$

1. (a) $I(\lambda,T) = \frac{2hc^2}{\lambda^5}\cdot\frac{1}{e^{\frac{hc}{\lambda kT}}-1}$, for a certain temperature, we can treat T as a constant, I as a function of λ

$$\frac{dI}{d\lambda} = -\frac{10hc^2}{\lambda^6}\cdot\frac{1}{e^{\frac{hc}{\lambda kT}}-1} + \frac{2hc^2}{\lambda^5}\cdot\frac{\frac{hc}{\lambda^2 kT}\cdot e^{\frac{hc}{\lambda kT}}}{(e^{\frac{hc}{\lambda kT}}-1)^2}$$

$$\frac{dI}{d\lambda} = \frac{2hc^2}{\lambda^6(e^{\frac{hc}{\lambda kT}}-1)}\cdot\left[-5+\frac{hc}{\lambda kT}\frac{e^{\frac{hc}{\lambda kT}}}{e^{\frac{hc}{\lambda kT}}-1}\right]$$

Since we know that $\left.\frac{dI(\omega)}{d\lambda}\right|_{\lambda=\lambda_m} = 0$, we have $-5+\frac{hc}{\lambda_m kT}\cdot\frac{e^{\frac{hc}{\lambda_m kT}}}{e^{\frac{hc}{\lambda_m kT}}-1} = 0$.

which is $\frac{hc}{\lambda_m kT} = 5(1-e^{-\frac{hc}{\lambda_m kT}})$

(b) Matlab code:

```
>>
clear;close all;
syms t;
eqn = t == 5*(1 - exp(-t));
s = vpasolve(eqn,t,[1 100])

s =

4.965114231744276303698759131322 9
```

$\Rightarrow$ the other solution: $\frac{hc}{\lambda kT} \approx 4.965$

(c) $\lambda_m = \frac{hc}{4.965\cdot kT}$

$\Rightarrow b = \frac{hc}{4.965\cdot k} \approx 2.898\times10^{-3}$

(d) $\lambda_m = \frac{b}{T}$

$\Rightarrow T = 5796\,K \approx 5523^\circ C$

(e) Result from Baidu: 5778 K in average, close to my result.

(f) $\lambda_m' = \frac{b}{T'}$

$\lambda_m' = \frac{2.898\times10^{-3}}{300} = 9.66\times10^{-6}\,m$

$f = \frac{c}{\lambda_m'} \approx 3.11\times10^{13}\,Hz$

$\Rightarrow$ the frequency of $3.11\times10^{13}\,Hz$ has the strongest emission at room temperature

2. a) Assume the wave function is $\psi(x)$

Boundary condition : $\psi(0) = \psi(L) = 0$

b) Assume $\psi(x) = A\sin kx + B\cos kx$

$\psi(0) = \psi(L) = 0$

$\Rightarrow B = 0. \quad A\sin kL = 0$

$\Rightarrow \begin{cases} k_n L = n\pi. \ (n = 0, 1, 2, \cdots) \\ k_n = \frac{2\pi}{\lambda_n} \end{cases}$

Possible wavelengths : $\lambda_n = \frac{2L}{n}$. $(n = 0, 1, 2, \cdots)$

c) corresponding $K_n = \frac{n\pi}{L}$ $(n = 0, 1, 2, \cdots)$

d) Energy levels : $E_n = \frac{(\hbar k_n)^2}{2m} = \frac{\hbar^2 n^2 \pi^2}{2mL^2}$ $(n = 0, 1, 2, \cdots)$

e) For electron, take $\hbar = 1.054 \times 10^{-34}$ J·s , $m = 9.11 \times 10^{-31}$ kg

$\lambda = 500$ nm , $f = \frac{1}{\lambda}$. $f = 2 \times 10^{6}$ Hz

$\begin{cases} E_0 = hf \\ E_0 = E_2 - E_1 \end{cases} \Rightarrow hf = \frac{3\hbar^2\pi^2}{2mL^2}$

$\Rightarrow L \approx 1.364 \times 10^{-10}$ m $= 1.364$ Å

3. (a) $P = mv$

$\Rightarrow P = 500\ kg \cdot m \cdot s^{-1}$

$P = \frac{h}{\lambda}$

$\Rightarrow$ De Broglie wavelength : $\lambda = 1.324 \times 10^{-36} m$

(b) It's not possible. To do the double slit interference experiment, the width of the slit should be narrower than $\lambda = 1.324 \times 10^{-36} m$, which is far less than atomic diameter ($10^{-10} m$)

(c) $E = 220\ eV \approx 3.524 \times 10^{-17} J$

$E = \frac{P^2}{2m}$

momentum : $P = 8.013 \times 10^{-24}\ kg \cdot m \cdot s^{-1}$

$P = \frac{h}{\lambda}$

De Broglie wavelength : $\lambda \approx 8.262 \times 10^{-11} m$

(d) $\lambda = 500nm = 5 \times 10^{-7} m$

$P = \frac{h}{\lambda}$

momentum : $P = 1.324 \times 10^{-27}\ kg \cdot m \cdot s^{-1}$

$E = \frac{P^2}{2m}$

$E \approx 9.621 \times 10^{-15} J$

$U = \frac{E}{q}$

Voltage : $U = 6.013 \times 10^{-6} V$

e) $E = 1.5kT$, take $T = 300K$

$P = \sqrt{2mE}$

momentum caused by random motion : $P \approx 1.064 \times 10^{-25}\ kg \cdot m \cdot s^{-1}$

which is larger than the momentum caused by acceleration, cannot do in room temperature.

4. $V(x) = \begin{cases} 0, & 0 < x < a \\ +\infty, & x \geq a \text{ or } x \leq 0 \end{cases}$

a) Assume wave function: $\psi(x)$

when $x \geq a$ or $x \leq 0$. $\psi(x) = 0$

when $0 < x < a$, $\frac{\partial^2 \psi(x)}{\partial x^2} + \frac{2mE}{\hbar^2} \cdot \psi(x) = 0$

$\Rightarrow \psi(x) = A \sin kx + B \cos kx$. where $k^2 = \frac{2mE}{\hbar^2}$

since $\psi(0^-) = \psi(0^+) = 0$. $\psi(a^-) = \psi(a^+) = 0$

we have $\psi(x) = A \sin ka$. $ka = n\pi$, $n = 0, 1, 2, \cdots$

$$\langle p_x \rangle = \int_0^a \psi^* \cdot \frac{\hbar}{i} \frac{\partial \psi}{\partial x} dx = -A^2 i\hbar \int_0^a \frac{e^{ikx} - e^{-ikx}}{2i} \cdot \frac{e^{ikx} + e^{-ikx}}{2i} \cdot ik \, dx$$

$$\langle p_x \rangle = -\frac{A^2 \hbar k}{2} \cdot \int_0^a \frac{e^{2ikx} - e^{-2ikx}}{2} dx = 0$$

b) $\int_0^a |\psi(x)|^2 dx = 1$

$$\Rightarrow \int_0^a A^2 \cdot \sin^2 kx \, dx = 1 \quad . \quad k = \frac{n\pi}{a}$$

$$A^2 \int_0^a \frac{1 - \cos \frac{2n\pi}{a} x}{2} dx = 1$$

normalization factor: $A = \sqrt{\frac{2}{a}}$.

c) $\psi(x) = \sqrt{\frac{2}{a}} \cdot \sin kx$

$$\langle x \rangle = \int_0^a \psi_{(x)}^* \cdot x \cdot \psi_{(x)} dx = \frac{2}{a} \int_0^a \sin^2 kx \cdot x \, dx$$

$$= \frac{1}{a} \cdot \int_0^a (1 - \cos 2kx) x \, dx$$

$$= \frac{a}{2} - \int_0^a \cos 2kx \cdot x \, dx$$

$$\langle x \rangle = \frac{a}{2}$$